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Q1.(Lambda Function)

Take input from the user

```
num = int(input("Enter a number: "))
```

Call the lambda function

```
print("Square using lambda function =", square(num))
```

Normal function using def

```
def square_def(x):
```

```
    return x * x
```

Call the normal function

```
print("Square using def function =", square_def(num))
```

Q2.(Lambda with Multiple Arguments)

```
multiply_three = lambda a, b, c: a * b *
```

```
num1 = int(input("Enter first number: "))
```

```
num2 = int(input("Enter second number: "))
```

```
num3 = int(input("Enter third number: "))
```

```
print("Multiplication =", multiply_three(num1, num2, num3))
```

Q3.(For Loop Revision)

a) Print all numbers from 1 to 25

```
print("Numbers from 1 to 25:")
```

```
for i in range(1, 26):
```

```
    print(i)
```

b) Print the sum of all numbers from 1 to 20

```
sum = 0
```

```
for i in range(1, 21):
```

```
    sum = sum + i
```

```
print("Sum of numbers from 1 to 20 =", sum)
```

c) Print the table of 5

```
print("Table of 5:")
```

```
for i in range(1, 11):
```

```
    print("5 x", i, "=", 5 * i)
```

Q4.(While Loop Revision)

```
total = 0
```

```
while True:
```

```
    num = int(input("Enter a positive number (0 or negative to stop): "))
```

```
    if num <= 0:
```

```
        break
```

```
    total += num
```

```
print("Total Sum =", total)
```

Q5.(Math Module)

```
import math

num = float(input("Enter a number: "))

print("Square Root =", math.sqrt(num))

if num >= 0 and num.is_integer():

    print("Factorial =", math.factorial(int(num)))

else:

    print("Factorial is only defined for positive integers.")

print("Ceiling Value =", math.ceil(num))

print("Floor Value =", math.floor(num))
```

Q6.(Random Module)

```
import random

print("5 Random Integers:")

for i in range(5):

    print(random.randint(1, 100))

print("Random Number between 50 and 150:", random.randint(50, 150))

print("Random Floating Point Number:", random.random())
```

Q7.(Import Methods)

```
import math
```

```
print("1. Power of 2^4 using math.pow():", math.pow(2, 4))

from math import sqrt

print("2. Square Root of 25 using sqrt():", sqrt(25))

import math as m

print("3. Factorial of 5 using m.factorial():", m.factorial(5))
```

Q8.(Custom Module)

```
def greet(name):

    print("Hello", name + ", Welcome to Python Class!")

def calculate_power(base, exp):

    return base ** exp
```

Q9.(name == "main")

```
def greet():

    print("Welcome to Python Programming!")

if __name__ == "__main__":

    greet()

    print("This program is running directly.")
```

Q10.(Mini Project - Combined Concepts)

```
import math

import random
```

```
square = lambda x: x * x

def calculate_power(base, exp):
    return math.pow(base, exp)

while True:
    print("\n===== Simple Math Utility =====")
    print("1. Square")
    print("2. Power")
    print("3. Random Number")
    print("4. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        num = int(input("Enter a number: "))
        print("Square =", square(num))
    elif choice == 2:
        base = int(input("Enter base: "))
        exp = int(input("Enter exponent: "))
        print("Power =", calculate_power(base, exp))
    elif choice == 3:
        print("Random Number =", random.randint(1, 100))
    elif choice == 4:
        print("Thank you! Program Ended.")
        break
    else:
        print("Invalid choice! Please try again.")
```

